Chen Tao

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Education

Carnegie Mellon University Pittsburgh, PA
Master of Science in Artificial Intelligence Engineering - Information Security Dec 2026

Nanyang Technological University Singapore
Bachelor or Engineering (Honours) in Electrical and Electronic Engineering Jun 2023
GPA: 4.79/5.00

Industrial Experiences

Agency for Science, Technology, and Research (A*STAR) Singapore
Research Engineer Aug 2023 – May 2025

- Constructed real-time automatic fact checker for fake news detection with **8%** accuracy improvement over Perplexity and Microsoft Bing
- Applied a test-time meta-learning framework in domain adaptation for generalizable deepfake detection
- Investigated effectiveness of membership inference attacks on zeroth-order optimization and identified data points with high privacy risks using canary insertion

Grab Singapore Headquarters Singapore
Software Engineer Intern Feb 2022 – May 2022

- Built keyword extraction models and applied topic modeling techniques to derive key insights from internal audit reports
- Designed automated Knime workflows that continually updated model to new data inputs

Singapore Power Group Singapore
AI Engineering Intern May 2021 – Jul 2021

- Developed an automated platform to manage technical interview environments, enabling streamlined testing and evaluation of job applicants
- Deployed service as an AI chatbot in Slack with intensive bash scripting and orchestration with Docker containers and Kubernetes

Research Experiences

Nanyang Technological University Singapore
Undergraduate Final Year Project Aug 2022 – May 2023

- Fine-tuned language models with contrastive learning and adapters, enhancing baseline performance by **7%** on the Patent Similarity dataset.
- Distilled knowledge from cross-encoder based models to bi-encoder based counterparts to reduce runtime overheads

Agency for Science, Technology, and Research (A*STAR) Singapore
Research Assistant Dec 2021 – May 2022

- Won **4th** place for Subtask A and **6th** place for Subtask B among over 400 participants on SemEval-2022 Task 5: Multimedia Automatic Misogyny Identification
- Outperformed baseline performance by **28%** by designing multitask multi-modal ensemble models to effectively fuse image and textual representations

Skills

Programming: Python, C/C++, JavaScript, PHP, SQL (MySQL, NoSQL), Bash, MATLAB
SWE: Docker, Kubernetes, OpenMPI, OpenFHE, Git, Linux
AI: PyTorch, TensorFlow, Keras, Hugging Face, Pandas, NumPy, Scikit-Learn, NLTK, SciPy, SpaCy, MMF, LangChain, RAG